

Postdoctoral Research Associate, Princeton University	
himani.sinhmar@princeton.edu himanisinhmar.github.io Google Scholar LinkedIn	
RESEARCH INTERESTS	My vision is to develop theoretically grounded and resource aware autonomy architectures that make robots safe, goal-directed, smoothly adaptive, and interpretable <i>by design</i> . My work draws on insights from dynamical systems, control theory, neuroscience, and collective behavior. I <i>design, analyze, and certify</i> methods for physical interactive autonomy that enables robots to act while deciding and decide while acting and <i>validate</i> them on robotic hardware.
POSITIONS HELD	Princeton University Postdoctoral Research Associate, Leonard Lab Advisor: Prof. Naomi Ehrich Leonard Sept. 2024–present
	Mitsubishi Electric Research Laboratories (MERL), Cambridge, MA PhD Research Intern, Control and Dynamical Systems Group 2023
	Indian Space Research Organization (ISRO), Bangalore, India Undergraduate Research Intern, Control and Dynamical Systems Group 2017
EDUCATION	Cornell University <i>Ph.D., Aerospace Engineering</i> Specialization in Robotics; Minor in Computer Science Doctoral Advisor: Prof. Hadas Kress-Gazit Aug. 2019–Aug. 2024
	Indian Institute of Technology (IIT Bombay) <i>B.Tech. and M.Tech., Aerospace Engineering</i> Specialization in Systems and Control; Minor in Physics Thesis Advisor: Prof. Srikant Sukumar 2014–2019
SELECTED HONORS AND AWARDS	Rising Star in Mechanical Engineering Stanford University, 2026
	Future Values Fellowship Princeton University, 2026
	Robotics Rising Star Princeton University, 2026
	Future Leader in Robotics and AI University of Maryland, 2025
	Caltech EAS Trailblazers Award Caltech, 2024 One of 7 early-career researchers in engineering and applied science (4% acceptance rate).
	Robotics: Science and Systems Pioneers RSS Conference, 2024 One of 30 leading early-career robotics researchers selected worldwide (15% acceptance rate).
	Hibbitt Postdoctoral Fellowship Brown University, offered and declined, 2024
	Graduate Resident Fellowship Cornell University, 2022–2024 8% acceptance rate.
	Institute Undergraduate Research Award IIT Bombay, 2018
MANUSCRIPTS IN PREPARATION	<i>Advised students are marked with [†] and equal contributions with *, when applicable.</i>
	[P1] Himani Sinhmar , Vaibhav Srivastava, and Naomi Ehrich Leonard. “Neuromorphic Decision Dynamics for Deadlock-free Safe Navigation in Congested Environments” <i>Science Robotics</i> (in prep.)
	[P2] Himani Sinhmar [*] , Antoun K. Salib ^{*,†} , and Naomi Ehrich Leonard. “Real-Time Inference of Cooperation for Path-Preserving Multi-Robot Social Navigation in Crowds.” <i>IEEE Transactions on Robotics</i> (in prep.)

	[P3] Himani Sinhmar , Giovanna Amorim [†] , Vaibhav Srivastava, and Naomi Ehrich Leonard. “Decentralized Neuromorphic Supervisory Control for Parsimonious Multi-Agent Persistent Environment Monitoring.” <i>International Journal of Robotics Research</i> (in prep.)
JOURNAL PUBLICATIONS	[J1] Marcus Greiff, Himani Sinhmar, Andreas Weiss, Kristofer Berntorp, and Stefano Di Cairano. “Invariant Set Planning for Quadrotors: Design, Analysis, and Experiments.” <i>IEEE Transactions on Control Systems Technology</i>, 33(2):449-462, 2024. [J2] Qingkun Liu*, Wei Wang*, Himani Sinhmar, Itay Griniasty, Jason Z. Kim, Jacob T. Pelster, Parag Chaudhari, Michael F. Reynolds, Michael C. Cao, David A. Muller, Alyssa B. Apsel, Nicholas L. Abbott, Hadas Kress-Gazit, Paul L. McEuen, and Itai Cohen. “Microscopic, Continuum, Compliant, and Electronically Configurable Metasheet Robots.” <i>Nature Materials</i>, 2024. [J3] Himani Sinhmar and Hadas Kress-Gazit. “Guaranteed Encapsulation of Targets with Unknown Motion by a Minimalist Robotic Swarm.” <i>IEEE Transactions on Robotics</i>, 40:816-830, 2023. [J4] Pallavi Sinha, Srikant Sukumar, and Himani Sinhmar. “Consensus of Networked Double Integrator Systems under Sensor Bias.” <i>International Journal of Adaptive Control and Signal Processing</i>, 37(1):278-297, 2022.
CONFERENCE PUBLICATIONS	[C1] Himani Sinhmar, Vaibhav Srivastava, and Naomi Ehrich Leonard. “Neuromorphic Realization of Best Response in Finite-Action Games.” To Appear in 2026 IEEE Conference on Decision and Control [C2] Ian Xul Belaustegui, Himani Sinhmar, Ling-Wei Kong, Andrew Michael Hein, and Naomi Ehrich Leonard. “A Continuous-Time and State-Space Relaxation of the Linear Threshold Model with Nonlinear Opinion Dynamics.” To Appear in 2026 IEEE Conference on Decision and Control [C3] Donggeon David Oh, Justin Lidard, Haimin Hu, Himani Sinhmar, Elle Lazarski, Deepak Gopinath, Emily S. Sumner, Jonathan A. DeCastro, Guy Rosman, Naomi Ehrich Leonard, and Jaime Fernández Fisac. “Safety with Agency: Human-Centered Safety Filter with Application to AI-Assisted Motorsports.” <i>Robotics: Science and Systems</i>, 2025. [C4] Himani Sinhmar and Hadas Kress-Gazit. “Multi-Source Encapsulation With Guaranteed Convergence Using Minimalist Robots.” <i>Distributed Autonomous Robotic Systems</i>, 2024. [C5] Himani Sinhmar, Marcus Greiff, and Stefano Di Cairano. “Practical and Safe Navigation Function Based Motion Planning of UAVs.” <i>IEEE International Conference on Robotics and Automation</i>, 2024. [C6] Himani Sinhmar and Hadas Kress-Gazit. “Decentralized Control of Minimalistic Robotic Swarms for Guaranteed Target Encapsulation.” <i>IEEE/RSJ International Conference on Intelligent Robots and Systems</i>, 2022. [C7] Himani Sinhmar and Srikant Sukumar. “Distributed Model Independent Algorithm for Spacecraft Synchronization under Relative Measurement Bias.” <i>5th CEAS Conference on Guidance, Navigation and Control</i>, 2019. [C8] Himani Sinhmar and Vinod Kumar. “Relative Autonomous Navigation Without Communication Between Spacecraft Using Line of Sight Measurements.” <i>IEEE/CSAA Guidance, Navigation and Control Conference</i>, 2018.

WORKSHOP PAPERS

- [W1] **Himani Sinhmar**, Itay Griniasty, Qingkun Liu, Wei Wang, Itai Cohen, and Hadas Kress-Gazit. “Design and Control of Microscopic Robot Sheet.” *ICRA Workshop*, 2024.
- [W2] **Himani Sinhmar**, Vaibhav Srivastava, Naomi Ehrich Leonard. “Neuromorphic Decision Dynamics and Control for Provably Safe and Deadlock-Free Multi-Agent Navigation in Congested Environments.” *IROS Workshop*, 2026.
- [W3] **Himani Sinhmar**, Antoun Salib, Naomi Ehrich Leonard. “Cooperation-Aware Neuromorphic Decision Dynamics for Path-Preserving Navigation in Heterogeneous Crowds.” *IROS Workshop*, 2026.

PATENTS AND DISSERTATION

- [PT1] Marcus Greiff, Stefano Di Cairano, and **Himani Sinhmar**. “[Motion Planning and Control with Multi-Stage Construction of Invariant Sets](#).” US Patent Application No. 18/589,946, 2025.
- [D1] **Himani Sinhmar**. “Control With Guarantees for Minimalist Robotic Swarms.” Ph.D. dissertation, Cornell University, 2024.

ADVISING AND MENTORSHIP

PhD Students Advised

2024–present

Giovanna Amorim (MAE Princeton)

Jeremy (JJ) Kuznetsov (MAE Princeton)

Leila Thompsky (Applied and Computational Mathematics Princeton)

Undergraduate Senior Thesis Advising

2026–2027 **Rebecca Shin** (MAE Princeton)

Thesis: *Integrated Planning and Navigation for Autonomous Agricultural Robots in Dynamic Environments*.

2026–2027 **Edric Zhang** (ECE Princeton)

2025–2026 **Antoun K. Salib** (ECE Princeton)

Thesis: *Path-Preserving Multi-Robot Navigation in Crowded Social Spaces with Online Cooperation Inference*.

Sigma Xi Honor Society Book Award for Outstanding Research

Professional Development

2025 Facilitating Mentor Training, CIMER Academic Mentor Training Workshop.

TEACHING

2021–2023 Autonomous Mobile Robots, Teaching Assistant, Cornell University.

2019 Dynamics and Control, Teaching Assistant, IIT Bombay.

2018 Spaceflight Mechanics, Teaching Assistant, IIT Bombay.

INVITED SEMINARS AND TALKS

Nov. 2025 Invited Talk at SIAM NNP Mini-Symposium on “Collective Behavior for Biologically Inspired Social Systems” Penn State University

Sept. 2025 Invited participant, NSF visioning workshop “Formalization of Human Factors into Computational Models,” sponsored by the Mind, Machine, and Motor Nexus program of NSF. Alexandria, VA

Apr. 2025 Invited seminar, Future leaders in robotics and AI, “Provably Scalable and Decentralized Control Design for Minimalist Robot Swarm” [\[Recording\]](#) University of Maryland

Nov. 2025 Invited Trailblazers seminar for CMS Caltech

	May 2024	Leonard Lab, <i>Princeton University</i> .	Princeton, NJ
	Feb. 2024	GRASP Spring Seminar, <i>University of Pennsylvania</i> .	Philadelphia, PA
	Feb. 2024	Automatic Coordination of Teams Lab, <i>Brown University</i> .	Providence, RI
	Jan. 2024	Biorobotics Lab, <i>Carnegie Mellon University</i> .	Pittsburgh, PA
	Jan. 2024	Distributed Robotics Lab, <i>MIT CSAIL</i> .	Cambridge, MA
	Jan. 2023	Collective Embodied Intelligence Lab, <i>Cornell University</i> .	Ithaca, NY
	May 2022	Guest lecture, Introduction to Computational Science and Engineering.	Cornell University
CONFERENCE AND POSTER PRESENTATIONS	2026	Invited poster, Learning Dynamics in Natural and Artificial Intelligence .	Harvard University
	2026	Invited poster, Diverse Intelligences Summit .	Princeton University
	2026	Poster, <i>Northeast Robotics Colloquium</i> .	Princeton University
	2025	Poster, <i>Northeast Robotics Colloquium</i> .	Cornell University
	2024	IEEE RAS Summer School on Multi-Robot Systems	Prague
	2024	Poster, <i>Robotics, Science and Systems Conference (RSS)</i> .	Delft University of Technology
	2023	Poster, <i>International Conference on Robotics and Automation (ICRA)</i> .	Japan
	2023	Poster, <i>Northeast Robotics Colloquium</i> .	Yale University
PROFESSIONAL SERVICE	Conference and Symposium Organization		
	2026	Organizing Committee, Princeton Robotics Symposium.	
	2025	Website and Publicity Chair, Robotics: Science and Systems Pioneers.	
	2025	Program Committee, 15th International Conference on Swarm Intelligence (ANTS 2026).	
	2022	Session Chair, Swarm Robotics, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).	
	2018	Session Chair, Navigation Technology, IEEE/CSAA Guidance, Navigation and Control Conference.	
	2016–2017	Chief Editor, Department Newsletter <i>Lift-Off</i> , IIT Bombay.	
Peer Review			
Journals		Conferences	
• <i>Science Advances</i>		• Robotics: Science and Systems	
• <i>IEEE Transactions on Robotics</i>		• IEEE Conference on Decision and Control	
• <i>IEEE Transactions on Automatic Control</i>		• IEEE International Conference on Robotics and Automation	
• <i>The International Journal of Robotics Research</i>		• IEEE/RSJ International Conference on Intelligent Robots and Systems	
• <i>Autonomous Robots</i>		• International Symposium on Distributed Autonomous Robotic Systems	
• <i>IEEE Robotics and Automation Letters</i>			